Practice Lab 3 Worksheet

Ques2. What is the time complexity of the following program?

(a) **O (n)**

(b) **O (n^2)**

(c) **O (n^3)**

(d) **O (N + M)**

(e) **O (N^2)**

(f) **O (n log n)**

(g) **O (log n)**

(h) **O (n log n)**

Ques3. For the functions, n^k and c^n, what is the asymptotic relationship between these functions? Assume that k >= 1 and c > 1 are constants.

$$n^k = O(c^k),\ k = 2 \text{ and } c = 2$$

Ques4. Decide whether these statements are True or False:

1. If f(n) = Θ(g(n)) and g(n) = Θ(h(n)), then h(n) = Θ(f(n)) **True**
2. If f(n) = O(g(n)) and g(n) = O(h(n)), then h(n) = Ω(f(n)) **True**
3. If f(n) = O(g(n)) and g(n) = O(f(n)) then f(n) = g(n) **False (for f(n) = n and g(n) = n+1)**
4. $\frac{n}{100}$ = Ω(n) **True** for c = 1/200